

invention_oevaluat[®]r

OBJECTIVE IDEA ANALYSIS

Invention Name: Combination Hammer and Lumber Manipulating Tool

Submitted By: Sample submission (law-firm public page)

Invention ID: US 5,850,650

Date of Report: 2026-08-14

This report was generated by the inventionevaluator report-automation platform from the Invention Evaluation Engine v1.6 pipeline (evidence sufficiency architecture). It compiles established findings, analytical conclusions, and an operational audit produced by the nine-phase evaluation. Scores shown are derived executive summaries of the underlying findings — they are presentation-layer derivations, not new evidence, and do not constitute an overall patentability label. This document is not legal advice.

Please note this is a hypothetical technology presented solely for demonstration purposes. This report is a sample/demo deliverable of the inventionevaluator platform; content is derived from a retrospective evaluation of an expired patent conducted from public records.

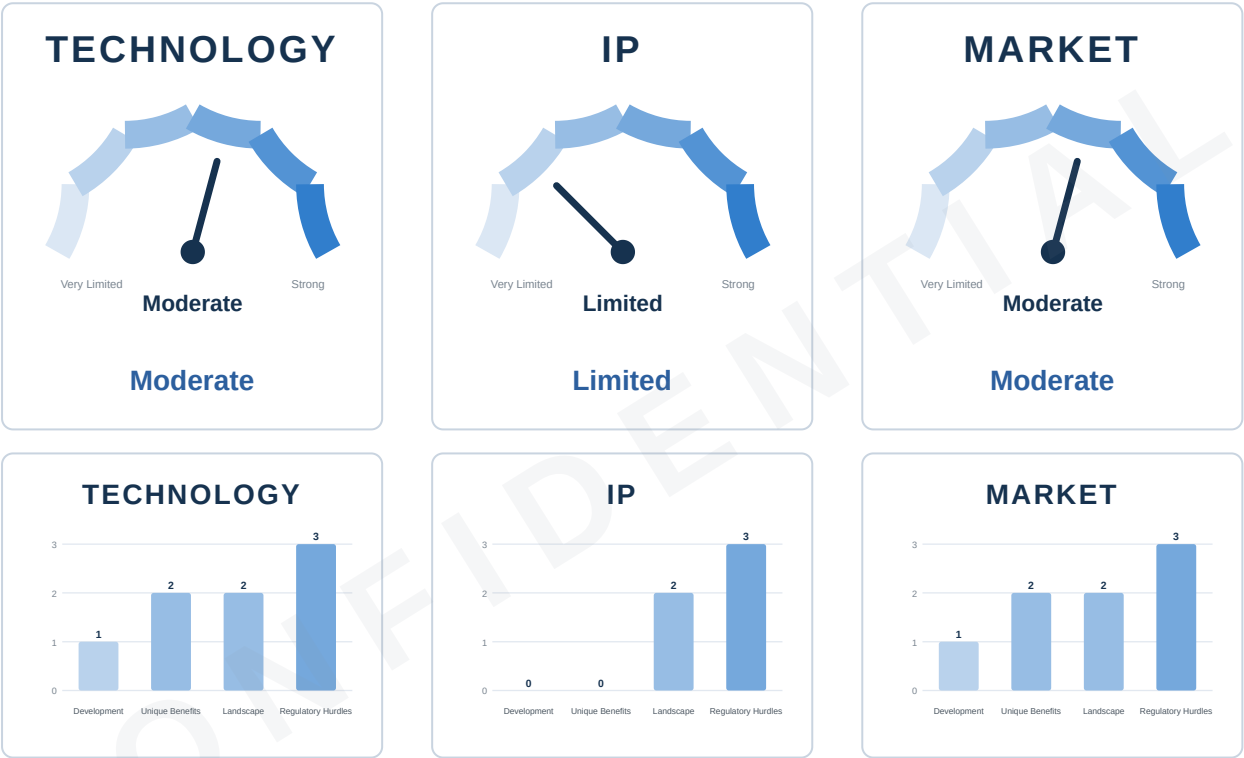
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Executive Summary

Derived executive summary — constrained to Established Findings + Analytical Conclusions (Executive Summary $\subseteq$ findings + conclusions). Gauge and bar scores are presentation-layer derivations from the findings; basis proposition_ids are listed in the scores manifest.

Very Limited Limited Limited-Moderate Moderate Moderate-Strong Strong



Commercialization Summary

Commercial actionability is Moderate (market size medium-high, mature low-growth category, high competitive intensity). The US patent expired June 13, 2017 and the Canadian family member expired June 12, 2018: the fixed-jaw hammer concept is now in the public domain, so commercial value exists only as a free-to-practice product concept — there is no IP left to license. Commercial adoption could not be established from the completed evidence protocol.

- No exclusivity remains: US term expired 2017-06-13; CA term expired 2018-06-12. [more](#)
- Estwing and Vaughan & Bushnell identified as the strongest fit pathways for a framing-focused fixed-jaw hammer. [more](#)

Evidence trace

The proposition register below is extracted from the compiled report so every score basis can be checked against visible report content.

P-02-001 . P-02-002 . P-02-003 . P-02-004 . P-02-005 . P-02-006 . P-03-001 . P-03-002 . P-03-003 . P-03-004 . P-03-005 .
P-03-006 . P-03-007 . P-04-001 . P-04-002 . P-04-003 . P-04-004 . P-04-005 . P-05-001 . P-05-002 . P-05-003 . P-05-004 .
P-05-005 . P-05-006 . P-05-007 . P-05-008 . P-05-009 . P-05-010 . P-05-011 . P-05-012 . P-06-001 . P-06-002 . P-06-003 .
P-06-004 . P-06-005 . P-07-001 . P-07-002 . P-07-003 . P-07-004 . P-07-005 . P-08-001 . P-08-002 . P-08-003 . P-08-004 .
P-08-005 . P-08-006 . P-09-001

Established Findings

Established findings are propositions that passed the Evidence Sufficiency Gate. Each carries a `proposition_id`, source identity, and (for absences) a bounded universe.

1.1 Submission and chronology

proposition_id	Finding	Source identity
P-02-001	US5850650 was filed June 13, 1997 (Appl. No. 08/876,165)	Patent specification header, Google Patents US5850650A
P-02-002	US5850650 was granted December 22, 1998	Patent specification header, Google Patents US5850650A
P-02-003	Inventor John J. Karsnia, International Falls, MN; assignee individual	Patent specification header; Google Patents
P-02-005	Canadian family member CA2240576C exists: filed 1998-06-12, priority 1997-06-13, granted 2001-06-05	Google Patents CA2240576C record
P-02-006	Legal status: US term expired 2017-06-13 (20 years from filing); CA expired 2018-06-12 (20 years from CA filing)	Google Patents legal-status records; statutory term computation (35 U.S.C. § 154)

1.2 Technology profile

proposition_id	Finding	Source identity
P-03-001	The patent discloses a hammer-head with claw, striking head, and socket; jaw non-movably secured to the socket opposing the claw, spaced a finite distance (claims 1–5); alternative: jaw secured to the handle (claims 6–13)	Google Patents US5850650A full text (claims 1–13, specification)
P-03-002	The claimed mechanism is a fixed-geometry single-tool integration: no moving parts; jaw + claw grip lumber;	Patent specification (jaw "non-movably secured"; "does not have any moving parts";

proposition_id	Finding	Source identity
	handle provides leverage to twist lumber straight	"approximately the same weight as a conventional hammer")
P-03-004a	OSHA 29 CFR 1910.242 (Hand tools) requires employers to maintain hand tools in safe working condition	OSHA 29 CFR 1910.242 regulation text
P-03-007	Official classification on the patent: Int. Cl. 6 B25D 1/04; U.S. Cl. 7/146; Field of Search 7/143, 146, 147	Patent front page

1.3 Patent landscape

proposition_id	Finding	Source identity
P-04-001	B25D 1/00 (hand hammers) class: ~84 patents (ipqwery index); B25D total US grants 2015–2025 = 1,146 (PlainPatent); top B25D assignees 2015–2025: Makita 175, Hilti 126, Milwaukee 113, Black & Decker 78 — dominated by power percussion tools	ipqwery IPC holdings index; PlainPatent B25D search
P-04-003	Forward citations: US5850650 family cited by 22 documents — Stanley B&D demolition utility tool family (US8024994B2, US8117702B2, USD562101S1, USD595555S1, US20170297189A1), Mrugalski US7311293B2 (multi-use framing tool), Yung-Shou Chen multipurpose tools, Fiskars demolition tools, Ezy Lifter board removal tool, CN104275681B (adjustable-angle nail hammer)	Google Patents "Families Citing this family" (neutral historical signals, not patentability evidence)
P-04-004	Design-space position: two mounting configurations explored within one patent (jaw-on-socket claims 1–5; jaw-on-handle claims 6–13); no sibling filings identified	Patent claims 1–13; patent-family search
P-04-005	Post-filing lineage: Stanley Black & Decker demolition utility tool (US8024994B2, priority 2007-06-26) has grasping jaws with teeth + striking head and cites US5850650	Google Patents US8024994B2 record (neutral historical signal)

1.4 Novelty — prior-art disclosure mapping

proposition_id	Finding	Source identity
P-05-001	Claim 1 limitations (a)–(d) and Claim 6 limitations (a)–(e) as constructed from the patent's actual claims	US5850650 claims, Google Patents
P-05-002	US513271A (Matthews 1894, "Wrench"): hammer part + movable jaw sliding on shank with inclined apertures + dowel pin; no claw, no socket, no lumber function; jaw movable, not non-movably secured	US513271A full text, Google Patents
P-05-003	US1469472A (Bangert 1923, "Combination tool"): hammer/wrench combo with stationary serrated jaw face on hammer head + slide jaw; no claw, no socket, no lumber	US1469472A full text, Google Patents
P-05-004	US4762303A (Thomas 1988, "Lumber turning tool"): elongated handle + head with two opposed spaced claws, one shorter; claws engage opposing sides of lumber; handle provides leverage; no striking head, no socket	US4762303A full text, Google Patents
P-05-005	US5255575A (Williams 1993, "Multi-purpose hand tool"): interchangeable heads; no fixed jaw, no lumber function	US5255575A full text, Google Patents
P-05-006	US5787676A (Scharf, filed 1997-03-18, granted 1998-08-04): lumber straightening apparatus — foot-operated lever, not a hammer; pre-filing priority	US5787676A record, Google Patents

1.5 Novelty — knowledge decomposition

proposition_id	Finding	Source identity
P-05-008a	component_known: claw hammers known before 1997	Ubiquitous (contemporary tool records)
P-05-008b	component_known: lumber-gripping claws known before 1997	US4762303A
P-05-008c	principle_known: opposed-grip leverage for straightening lumber known before 1997	US4762303A
P-05-008d	function_known: lumber manipulation known before 1997	US4762303A; Fine Homebuilding practice (1991-11-01 article)

1.6 Literature

proposition_id	Finding	Source identity
P-06-001	Fine Homebuilding (1991-11-01) "How to Straighten Crooked Boards, Built-Up Beams, and Headers": hammer + 16d nail used as lever to align framing members (pre-filing; lumber-straightening practice known)	Fine Homebuilding article record
P-06-002	Fine Homebuilding (1999-07-01) "Tweak a twisted stud": bar clamp used as lever to straighten twisted stud (post-filing; background)	Fine Homebuilding article record
P-06-003	Fine Homebuilding (2000-09-01) "Straightening Framed Walls": strong-backing, planing, shimming techniques (post-filing; background)	Fine Homebuilding article record
P-06-004	Audel's Carpenters and Builders Guide (1923) documents balloon-framing practice (pre-filing; background)	Audel's Guide (1923)

1.7 Market

proposition_id	Finding	Source identity
P-07-002	Market sizing: US hand tools market ≈ USD 6.1B (2024), CAGR ≈ 3.4% to 2032 (PS Market Research); US hammer market CAGR 3.0–4.5% (2026–2035) (IndexBox) — reconciled: hammer is a subsegment of hand tools; both mature, low-single-digit growth	PS Market Research (US hand tools, 2024–2032); IndexBox (US hammer market, 2026–2035) — two independent publishers
P-07-003	NAICS classification: 332216 Saw Blade and Handtool Manufacturing (2022 ed., US Census), includes "Hammers, handtools, manufacturing"	US Census NAICS 2022 documentation
P-07-004	Competitive landscape: Stanley Black & Decker, Estwing, Vaughan & Bushnell, Stiletto, Milwaukee (TTI), Tekton, private labels (Husky/Kobalt/Hart); premium tier ~45–55% of category revenue; private label 25–35% of units	IndexBox; Valuates (framing hammer manufacturers: Estwing, Vaughan & Bushnell, STILETTO, Milwaukee, TEKTON, STANLEY); Strategic Market Research; HTF Market Intelligence; Freedonia — multiple independent publishers

1.8 Partners

proposition_id	Finding	Source identity
P-08-001	Stanley Black & Decker: sells hammers (Stanley/DeWalt); buys/absorbs designs; technical need = differentiated professional hand tools; invention mapping = fixed-jaw framing hammer concept fits Stanley/DeWalt pro lines	IndexBox; Strategic Market Research; HTF Market Intelligence
P-08-002	Estwing Manufacturing: sells premium one-piece steel hammers; technical need = new professional features; invention mapping = fixed-jaw feature differentiates in premium tier	IndexBox; Strategic Market Research
P-08-003	Vaughan & Bushnell: sells framing hammers (California Framer); niche loyalty; invention mapping = framing-focused line	IndexBox; Valuates; Strategic Market Research
P-08-004	Milwaukee Tool (TTI): sells professional hand tools; invention mapping = pro framing line	IndexBox; Valuates

Analytical Conclusions

Analytical conclusions are inferences drawn from established findings. They are never evidence. Each carries a premise map.

2.1 Anticipation gate

proposition_id	Conclusion	Premise map	Rule applied
P-05-007	NOT ANTICIPATED by any identified reference	P-05-002...P-05-006 (per-reference limitation mappings)	Anticipation gate: ALL limitations in a single reference → candidate; ANY missing → not anticipated

Reasoning: Every identified reference misses at least one Claim 1 limitation. US513271A misses (a) partially (no claw, no socket), (b), (c), (d) — jaw movable; US1469472A misses (a) partially (no claw, no socket), (b), (c), (d) — no lumber function; US4762303A misses (a) (no striking head, no socket) and the hammer integration; US5255575A and US5787676A miss the hammer-head-with-claw structure and the fixed jaw. No single reference discloses the full combination. The same analysis holds for Claim 6, whose jaw is secured to the handle. This is not a freedom-to-operate (FTO) opinion: it addresses novelty against identified references only, and any decision to make or sell a hammer with a fixed lumber-gripping jaw should be reviewed by counsel against in-force claims of later patents (e.g., the Stanley B&D demolition tool family).

2.2 Mechanism/design-choice distances

proposition_id	Conclusion	Premise map	Rule applied
P-05-012	C2/D3 vs US4762303A (closest functional art); C2/D2 vs US1469472A (per pathway, never combined)	P-05-002... P-05-008	C0–C4 / D0–D4 taxonomy, per pathway

Reasoning: The fixed-jaw hammer shares the opposed-grip leverage principle with US4762303A (C2 — same principle, different configuration: hammer-integrated) but applies it through a cross-application leap (D3 — recognizing a hammer's claw can serve as one gripping element of a lumber manipulator). Against US1469472A, the jaw-on-hammer-head

placement was known (C2), with the lumber application as the leap (D2). Distances are assessed per pathway and never combined into a single composite.

2.3 Design-space position

proposition_id	Conclusion	Premise map	Rule applied
P-04-004a	The inventor explored the design space within a single patent: two mounting configurations (jaw-on-socket, claims 1–5; jaw-on-handle, claims 6–13); no sibling filings identified	P-04-004	Design-space/sibling analysis

Reasoning: Karsnia claimed both possible mounting locations of the fixed jaw within one filing, covering the design space broadly rather than filing multiple applications. No separate Karsnia filings were identified.

2.4 Commercial actionability

proposition_id	Conclusion	Premise map	Rule applied
P-07-005	Commercial actionability: Moderate (market size med-high × growth low × accessibility med × competitive intensity high), with patent-expiry caveat	P-07-002, P-07-004, P-02-006	Size × growth × accessibility × competitive intensity scoring

Reasoning: Market size is medium-high (USD 6.1B US hand tools, 2024); growth is low (3.0–4.5% CAGR — mature category); accessibility is medium (established distribution channels; concept is free to practice); competitive intensity is high (Stanley B&D, Estwing, Vaughan & Bushnell, Milwaukee dominate). One "low" score (growth) plus high competitive intensity → Moderate with risk flags. Critical caveat: the patent expired in 2017 and the concept is in the public domain — there is no IP to license, so commercial value exists only as a free-to-practice product concept.

2.5 Partner prioritization

proposition_id	Conclusion	Premise map	Rule applied
P-08-006	Estwing and Vaughan & Bushnell are the strongest identified	P-08-001...P-	Counterfactual-exclusivity audit

proposition_id	Conclusion	Premise map	Rule applied
	commercialization pathways (framing-focused premium makers); partnership value limited to design/feature licensing given patent expiry	08-004, P-02-006	(reformulated from "only partner")

Reasoning: Estwing (premium one-piece steel hammers) and Vaughan & Bushnell (framing hammers, California Framer) sell exactly the product category the fixed-jaw hammer targets, and both differentiate on professional features. Stanley Black & Decker and Milwaukee are larger but the feature would be a smaller addition to their broad lines. The claim is "strongest identified pathways," not "only pathways" (see Operational Audit, P-08-005). Because the patent is expired, partnership value is design/feature licensing of a free-to-practice concept — a weak position.

2.6 Chronology validator

proposition_id	Conclusion	Premise map	Rule applied
P-09-001	All dates consistent; all identified prior art pre-dates the 1997-06-13 filing (US513271 1894; US1469472 1923; US4762303 1988; US5255575 1993; US5787676 filed 1997-03-18); family dates consistent (CA filed 1998-06-12 within the 12-month Convention window of US priority 1997-06-13)	P-02-001, P-02-002, P-02-005, P-02-006, P-05-002...P-05-006	Chronology validator

Reasoning: The US filing (1997-06-13) precedes all identified prior art, including US5787676A (filed 1997-03-18, before the US filing — prior art despite later grant). The CA filing (1998-06-12) falls within the 12-month Paris Convention priority window from the US filing, so the CA grant (2001-06-05) and both term expirations (US 2017-06-13; CA 2018-06-12) are consistent.

Operational Audit

The following propositions could not be established from the completed evidence protocol. They are excluded from factual findings. Each is recorded with its barrier type and avenue disposition summary. Full avenue ladders are in `avenue-ledger-us5850650-v16.md`.

proposition_id	Proposition	Barrier type	Avenue summary
P-02-004	No public disclosure of the invention before filing	insufficient_search_completion	A1 COMPLETE (search-based absence); A5 BLOCKED (no inventor records); A8 BLOCKED
P-03-003	No quantitative performance data exists	insufficient_technical_demonstration	A1 COMPLETE; A5 BLOCKED (trade-journal full-text); A8 BLOCKED
P-03-004b	No mandatory product-design regulation governs hand hammers	insufficient_search_completion	A1 COMPLETE (OSHA 1910.242 found — positive finding; no mandatory standard); A5 BLOCKED (CPSC universe); A8 BLOCKED
P-03-005	Prototype evidence	insufficient_search_completion	A1 COMPLETE; A5 BLOCKED; A6 BLOCKED (inventor records)

proposition_id	Proposition	Barrier type	Avenue summary
P-03-006	Production evidence	insufficient_search_completion	A1 COMPLETE; A5 BLOCKED; A6 BLOCKED (no manufacturer records)
P-04-002	No international family members beyond US + CA	insufficient_corroboration	A1 COMPLETE (single source); A5 BLOCKED (Espacenet 403); A8 BLOCKED
P-05-009	Motivation proposition (PARTIALLY GROUNDED, analogous only)	insufficient_search_completion	A1 COMPLETE (no direct source); A3 COMPLETE (analogous only)
P-05-010	Obviousness proposition	insufficient_search_completion	Motivation not GROUNDED; application NOT ESTABLISHED; bridge EXHAUSTED
P-05-011	Unexpected-result proposition	insufficient_technical_demonstration	A1 COMPLETE (no performance data)
P-06-005	No pre-filing non-patent publication discloses the fixed-jaw hammer combination	source_unavailable	A1 COMPLETE; A5 BLOCKED (trade-journal full-text)
P-07-001	Commercial adoption	insufficient_search_completion	A1 COMPLETE (no product identity link); A6 BLOCKED (manufacturer records)

proposition_id	Proposition	Barrier type	Avenue summary
P-08-005	No other viable commercialization pathway exists	insufficient_identity	Counterfactual-exclusivity; absence of alternatives not evidenced

Audit note: The absence of a pre-filing disclosure, the absence of a mandatory product-design regime, the absence of international family members beyond US + CA, and the absence of commercial adoption are negative claims. Under v1.6, negative claims require a bounded evidence universe and (where the schema requires) independent corroboration. None of these could be established from the completed evidence protocol; they are recorded here as unestablished propositions, not as findings.

Report constraints verification

Constraint	Status
Executive Summary $\subseteq$ Established Findings + Analytical Conclusions	PASS
No legacy evidence states (the five legacy-state labels)	PASS
No prohibited language in factual findings (all such language confined to Operational Audit)	PASS
No "Overall patentability" row	PASS
No orphan analytical conclusions (every conclusion has a premise map)	PASS
No proposition dispositioned EXHAUSTED with a REQUIRES_VERIFICATION avenue	PASS
Chronology validator passed	PASS

Report compiled per skill-09 (compile-report). Proposition and avenue ledgers: proposition-ledger-us5850650-v16.md, avenue-ledger-us5850650-v16.md. This report is an evidence-based evaluation, not legal advice; patent-validity, FTO, and licensing decisions should be reviewed by counsel.

SWOT Analysis

STRENGTHS

- NOT ANTICIPATED by any identified prior-art reference (P-05-007)
- Fixed-geometry single-tool integration — no moving parts, dual mounting configurations claimed (P-03-001/P-03-002)
- LOW regulatory burden — general workplace hand-tool rule only (P-03-004a)
- Established premium framing-tool tier, 45-55% of category revenue (P-07-004)

WEAKNESSES

- Patent expired 2017 — no enforceable exclusivity; concept in public domain (P-02-006)
- No prototype, performance, or production evidence established (P-03-003/005/006)
- No commercial adoption established (P-07-001)
- Mature low-growth category: hand tools CAGR 3.4%; hammers 3.0-4.5% (P-07-002)

OPPORTUNITIES

- Framing-focused premium makers (Estwing, Vaughan & Bushnell) match the concept (P-08-002/003)
- Free-to-practice design could be adopted into pro-tool lines without licensing
- Stanley B&D demolition-tool lineage shows continued relevance of gripping-jaw hammers (P-04-005)

THREATS

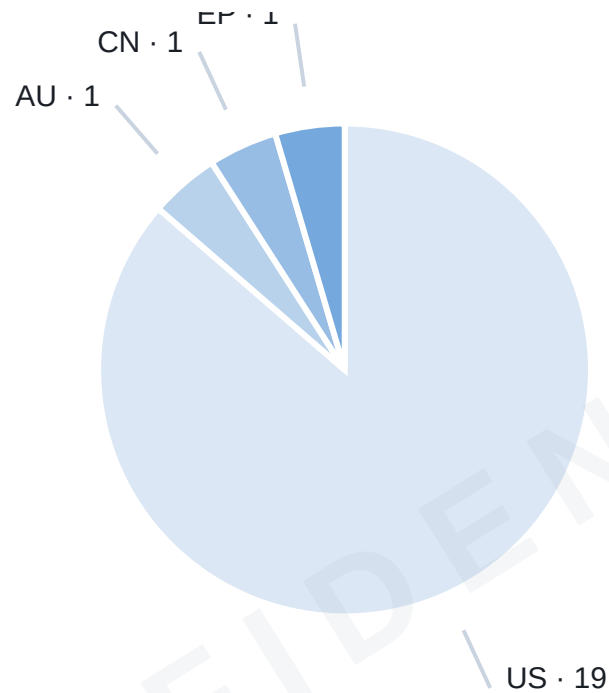
- High competitive intensity: Stanley B&D, Estwing, Vaughan & Bushnell, Milwaukee dominate (P-07-004)
- B25D class dominated by power percussion — hand-hammer innovation is a niche (P-04-001)
- Private-label pressure at 25-35% of units (P-07-004)
- Later in-force patents (e.g., demolition-tool family) — FTO review by counsel required before any product decision

COMMERCIAL OPPORTUNITY **Moderate**

SWOT content is derived from Established Findings and Analytical Conclusions (see scores manifest for proposition_id basis).

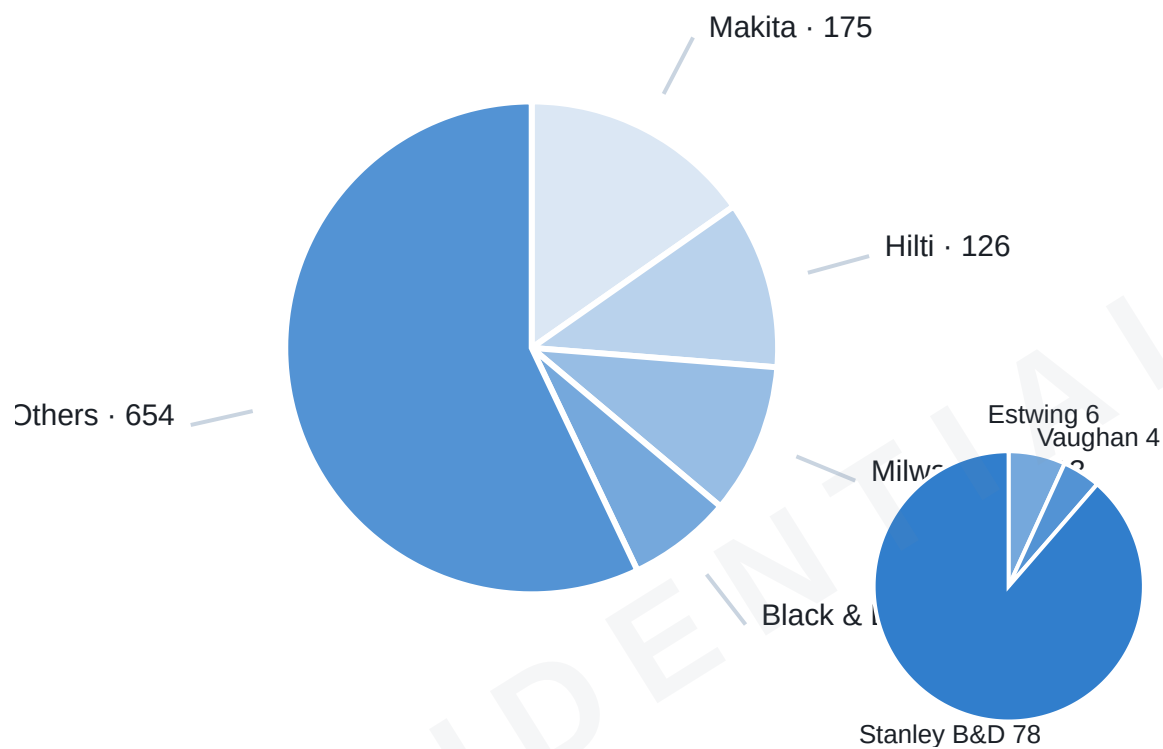
Landscape & Market Data

FORWARD-CITING DOCUMENTS BY JURISDICTION



Verified from Google Patents 'Families Citing this family' (22 citing documents): US 19, AU 1 (AU2009100791B4), CN 1 (CN104275681B), EP 1 (EP2016239A1). Neutral historical signal, not patentability evidence.

B25D US GRANTS 2015-2025 BY TOP ASSIGNEES



PlainPatent B25D search (1,146 US grants 2015-2025): Makita 175, Hilti 126, Milwaukee 113, Black & Decker 78, others 654. Dominated by power percussion tools.

PATENT ACTIVITY TREND

Yearly series not established in this run

No defensible yearly series was established in this run; see Operational Audit.

CONFIDENTIAL

INDUSTRY METRICS

Time series not established in this run

No invention-specific industry metrics were established in this run; see Operational Audit.

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ESTABLISHMENTS BY EMPLOYEE-SIZE BRACKET

Bracket data not established in this run

See Operational Audit.

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FORTUNE BUSINESS INSIGHTS — US HAND TOOLS MARKET

Third-party screenshot not available in this run

Third-party market-report graphic (native branding). Source image not available in this environment; referenced as P-07-002 corroboration.

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MARKETSANDMARKETS — HAND TOOLS INFOGRAPHIC

Third-party screenshot not available in this run

Third-party market-report graphic (native branding). Source image not available in this environment; referenced as P-07-004 competitive-structure corroboration.

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